ACS Technologies Limited

Technology & Innovations Division

Plan Evaluation Report

DroneArjuna — Closing the Phase 3–6 Gaps

Aug–Sep 2026 Project Plan Review & Gap Analysis Evaluation

Document Reference	ACSTL/DoTI/DAPE/DA/2026/001
Evaluated Document	DA_gaps.pdf — DroneArjuna Project Plan Aug–Sep 2026 (11 pp, 07-Aug-2026)
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Reviewed By	Director, Technology & Innovations
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1. Executive Summary

The DroneArjuna Aug–Sep 2026 project plan (DA-Project-Plan-Aug-Sep-2026, authored by Chandra Sekhar) is evaluated here against the original SRS, architecture, and tech-stack decisions established in prior ACSTL project sessions, and against ACSTL's internal delivery standards.

The plan is substantially stronger than a typical gap-closure document produced under operational constraints. Chandra Sekhar has correctly identified 14 discrete gaps from the as-built status report, costed them in person-days against a realistic two-window capacity model, sequenced them against an explicit dependency graph, and called out the single highest-uncertainty item (G5 — YOLOv8) with an honest risk statement. The plan is safe to approve.

The delays acknowledged in this plan are operational in origin — Vaishnavi's reduced availability in Window A — not attributable to poor planning, wrong tooling choices, or architectural missteps. The team is adhering to the original tech stack and architecture. The plan is a competent response to a capacity constraint.

However, three structural omissions reduce the plan's actionability and create sign-off risk at the 30-Sep gate:

- No Definition of Done exists for any of the 14 gaps. Milestones are stated but no per-gap pass/fail criteria are defined, making the 30-Sep close-out gate subjective.
- The YOLOv8 integration (G5) carries an 8 PD estimate without a pretrained model baseline evaluation. If the model performs poorly on drone-specific imagery, the impact cascades to G6, G7, G9, and the 14-Sep milestone.
- There is no rollback or contingency plan for milestone failures. If 21-Aug slips, the plan has no documented recovery path.

1.1 Coverage scorecard

Evaluation area	Assessment	Rating
Plan structure & sequencing logic	Strong	✓ Well-formed
Capacity model accuracy	Strong	✓ Well-formed
Gap identification completeness	Good	~ Minor gaps
Effort estimation realism	Adequate	~ Needs caveats
Risk register depth	Shallow	⚠ Thin
YOLOv8 / ML integration specification	Thin	⚠ Needs work

Evaluation area	Assessment	Rating
Definition of Done per gap	Absent	✗ Missing
Rollback & contingency planning	Absent	✗ Missing
Post-30-Sep acceptance criteria	Absent	✗ Missing
Overall plan maturity	~72%	**Approve with notes**

2. Plan Strengths — What the Team Got Right

This section records specific decisions in the plan that deserve recognition, both because they are correct and because they should not be revised under time pressure.

2.1 Capacity model is honest and realistic

The plan does not paper over Vaishnavi's Window A constraint. It explicitly models her at 25% (4 PD over 16 working days), assigns her only tasks that are directly paired with Chandra Sekhar's parallel work, and explains the rationale: avoiding the context-switch overhead a standalone task at 25% availability would carry. This is sound workforce planning. The note about elapsed calendar time being four times the PD estimate at 25% availability is a useful reminder for milestone tracking.

2.2 Dependency graph is correctly formed

The G4 → G5/G6 → G7 dependency chain is right. Background worker infrastructure (G4) must be stable before inference (G5) or video extraction (G6) can be wired. G1 being independent of all backend work means Indra is never blocked. G3's backend/frontend split across windows is well-structured. The decision to schedule CI pipeline (G11) mid-plan rather than first — once G1 and G4 exist as real code paths — is explicitly justified and correct.

2.3 YOLOv8 risk is called out directly

Naming G5 as the highest-uncertainty item and budgeting a 3–5 PD overage buffer (drawn from G13 if needed) is the right approach. Most junior-authored plans bury this risk. The plan surfaces it clearly in Section 08.

2.4 Window B frontend is slightly over capacity — correctly flagged

Window B frontend loads ~23 PD against 22 PD capacity (G2: 5, G9: 8, G12: 5, G14: 0, G3 frontend: 3, G11 frontend: 2). The 1 PD overage is small but the plan's sequencing of G14 (finish) and G3 (frontend) at the end of Window B means late-window tasks are the natural cut candidates if the earlier items slip. This is implicit priority management — it should be made explicit.

3. Gap Register Evaluation — Fourteen Gaps Annotated

The table below evaluates each of the 14 gaps from the plan's Section 02 Gap Register. Columns reproduce the plan's own data; the Evaluation notes column is the Director's assessment.

Ref	Gap	Track	PD	Evaluation notes
G1	3D visualisation scaffold (Cesium/terrain)	Frontend	10	none
G2	Telemetry replay player (3D + timeline)	Frontend	5	G1
G3	Drone Inventory knowledge-base workflow	Both	8	none
G4	Analyst background worker (RabbitMQ)	Backend	6	none
G5	YOLOv8 object-detection inference	Backend	8	G4
G6	Video-frame extraction pipeline (ffmpeg)	Backend	5	G4
G7	Change-detection module	Backend	6	G5,G6
G8	MinIO object storage wiring	Backend	4	none
G9	Analyst frontend workspace	Frontend	8	G4,G5
G10	Automated test coverage ≥80%	Both	6	ongoing
G11	CI pipeline (lint, type-check, test, build)	Both	4	none
G12	Accessibility pass (WCAG across 5 workspaces)	Frontend	5	none
G13	Load testing (10+ drone telemetry + API)	Backend	4	G4–G8 stable
G14	Frontend automated test suite (zero→first pass)	Frontend	6	none

4. Structural Gaps in the Plan Document

These are gaps in the plan itself — not in the product — that should be resolved before the plan is formally signed off.

4.1 No Definition of Done for any gap

The single most important missing element. The 30-Sep milestone states "all Phase 3–5 gaps closed or explicitly deferred," but there is no specification of what "closed" means for any individual gap. Without a DoD, the close-out review becomes a negotiation rather than a verification.

> **Example — G5 currently:** YOLOv8 inference producing real detection results on test imagery (04 Sep milestone).

> **Required DoD for G5:** YOLOv8 inference endpoint returns bounding-box results with confidence scores for ≥90% of a 30-frame reference test set. At least one detection result is persisted to the detection-result table. Inference latency on reference hardware is measured and recorded.

A half-day session with Chandra Sekhar and Indra to write 2–3 DoD statements per gap would significantly de-risk the 30-Sep gate.

4.2 YOLOv8 baseline evaluation not scheduled

The plan allocates Sep 1–4 and Sep 7–9 to G5 (8 PD total). This assumes pretrained YOLOv8 weights are adequate for drone-collected imagery. Drone imagery has distinct characteristics from the COCO/ImageNet training distribution: high-altitude perspective, small object density, variable lighting, compressed video artefacts. There is no scheduled activity to validate model baseline performance before the full 8 PD is committed.

The recommended remediation is to schedule a one-day spike (Aug 12, PD drawn from Chandra Sekhar's G4 buffer or the Aug 31 catch-up day) to run YOLOv8n and YOLOv8s against 20–30 representative frames. If mAP50 is below 0.35, the 8 PD estimate for G5 is materially wrong and the plan needs revision before Sep 1.

4.3 ffmpeg is not in the original tech stack

G6 specifies "ffmpeg-based frame sampling from uploaded video." The original DroneArjuna tech stack and architecture document does not list ffmpeg as a dependency. This is not a plan-breaking issue — ffmpeg is a natural fit for the task and the Python bindings (ffmpeg-python, av) are mature. However, it needs to be:

- Added to requirements.txt with a pinned version.
- Explicitly added to the Dockerfile/container specification.
- Noted as a tech stack amendment in the design decisions addendum (DA-DDD-001) that Vaishnavi is authoring in Window A.

4.4 No rollback or contingency procedure

The plan identifies what could move it (Section 08 — Risks & Notes) but does not define what happens when a milestone is missed. Six dated milestones span 10 weeks. If the 21-Aug milestone slips by more than 3 days, it compresses the remaining Window A work and reduces the Aug 31 review's utility as a genuine gate.

A lightweight contingency note per milestone is sufficient — one sentence covering: (a) what is deferred if the milestone fails, (b) who decides, and (c) whether the 30-Sep commitment is affected. This does not need to be elaborate.

4.5 Post-30-Sep acceptance gate is undefined

The plan ends at 30-Sep with "≥80% verified coverage, accessibility pass complete, all Phase 3–5 gaps closed or explicitly deferred." There is no definition of the acceptance process: who reviews the deferred items list, what constitutes an acceptable deferral reason, and who signs off that the product is ready for the next engagement phase (field trials, demo, or procurement documentation).

Given that DroneArjuna is targeting Indian Army procurement, the 30-Sep state of the product will form the basis of any demonstration or technical evaluation. The acceptance criteria should reference that context.

5. Milestone Table — Suggested Acceptance Criteria

The plan's Section 07 milestones are reproduced below with suggested acceptance criteria added. These are proposed by the Director for discussion with Chandra Sekhar before 10 August.

Date	Milestone	Gaps closed	Suggested acceptance criterion
21 Aug	Analyst job worker + MinIO end-to-end (backend)	G4 ✓, G8 ✓	Vaishnavi G4 schema review complete; presigned URL path tested

Date	Milestone	Gaps closed	Suggested acceptance criterion
31 Aug	3D visualisation scaffold usable; Window A close-out	G1 ✓, G14 partial	Cesium terrain tiles loading in Fly workspace; first Vitest suite passing
04 Sep	YOLOv8 inference producing real detections on test imagery	G5 partial	Confidence threshold defined; at least 10 labelled test frames validated
14 Sep	Change-detection + video pipeline functional; Analyst UI end-to-end	G6 ✓, G7 ✓, G9 partial	ffmpeg pipeline tested on 3 video formats; UI job-launch flow complete
21 Sep	CI pipeline green (backend + frontend); load-test report delivered	G11 ✓, G13 ✓	GitHub Actions passing on main; 10-drone load test meets original NFR latency
30 Sep	≥80% verified coverage; accessibility pass complete; all gaps closed or deferred	G10 ✓, G12 ✓, G14 ✓	Director sign-off on deferred items list; coverage report committed to repo

6. Required Actions

Actions A1 and A2 are prerequisites for Director sign-off on the plan. Actions A3–A10 should be completed within the first week of Window A.

Ref	Action	Owner	By	Priority	Rationale
A1	Define DoD per gap (G1–G14)	Chandra Sekhar + Indra	Before 10 Aug	High	Plan has milestones but no per-gap acceptance criteria. Each gap needs 2–3 measurable pass/fail statements.
A2	Establish YOLOv8 baseline on sample drone imagery	Chandra Sekhar	12 Aug	Critical	Run pretrained YOLOv8n/s against 20–30 representative frames before committing 8 PD to G5. If mAP < 0.4, re-plan G5 now, not in late September.
A3	Pin ffmpeg version; decide container packaging for G6	Chandra Sekhar	Before G6 start (Sep 10)	High	ffmpeg not in original stack. Add to requirements.txt/Dockerfile and document the version.
A4	Define job-state schema for G4 (PostgreSQL table vs RabbitMQ-native)	Chandra Sekhar + Vaishnavi	13 Aug (G4 design day)	High	Vaishnavi's G4 unit tests depend on a stable schema. Decision must precede her Aug 10–14 work.
A5	Add rollback procedure for each milestone	Chandra Sekhar	Before 21 Aug milestone	Medium	If the 21 Aug milestone (Analyst worker + object storage end-to-end) fails, what is the recovery action? Plan is silent on this.
A6	Set WCAG standard explicitly (2.1 AA)	Indra	Before G12 start (Sep 18)	Medium	Ambiguity on WCAG level creates scope risk. 2.1 AA is the correct baseline for a defence GCS.
A7	Specify polling interval for G9 Analyst frontend	Indra + Chandra Sekhar	Before G9 start (Sep 7)	Medium	1s vs 5s polling has material UX and backend load implications. Agree before implementation.

Ref	Action	Owner	By	Priority	Rationale
A8	Obtain current test coverage baseline	Chandra Sekhar + Vaishnavi	10 Aug (first day)	**High**	G10 targets ≥80% but no current baseline is documented. Run pytest-cov on Day 1 and record the number.
A9	Add post-30-Sep acceptance gate criteria to plan	Director review	Before 10 Aug sign-off	**High**	The plan closes gaps or defers them. What are the criteria for a gap to be formally "closed"? Who signs off?
A10	Plan for G13 load test against original NFR (sub-100ms C2 latency)	Chandra Sekhar	Before Sep 18	**Medium**	Load test should validate the original SRS NFR, not just confirm 10-drone survival. Add the specific latency target to the G13 test plan.

7. Director's Recommendation

The plan is approved subject to Actions A1 and A2 being completed before or on 10 August 2026. The team's approach to the operational constraint (Vaishnavi's availability) is sound and should not be second-guessed. The dependency sequencing and capacity model are correct.

The three items to watch across the plan period:

- G5 (YOLOv8) — the single item most likely to require re-planning. The Aug 12 baseline spike is not optional.
- Window B frontend load (~23 PD against 22 PD capacity) — monitor at the 14-Sep milestone; G12 (accessibility) and G3 frontend are the natural candidates for partial deferral if Indra is behind.
- 30-Sep gate — the Director should schedule a half-day review on 29-Sep to assess the deferred items list before the formal sign-off on the 30th.

> **Approved with conditions:** Complete A1 (Definition of Done) and A2 (YOLOv8 baseline spike) before 10 August. Proceed to Window A as planned. Report status at the 21-Aug milestone.

8. Appendix — Context References

The following prior ACSTL documents informed this evaluation:

- DA-Implementation-Status.pdf, 5 Aug 2026 — as-built status baseline referenced in the plan.
- DroneArjuna SRS v1.0 — five-module GCS architecture: Drone Control, Drone Master, Drone Inventory, Drone Flight, Drone Analyst. Python/FastAPI backend, React/Vite/Cesium.js frontend, PostgreSQL, MAVLink.

- ACSTL Technology Roadmap 2026–2028 — DroneArjuna positioned for Army procurement targeting imported GCS replacement.
- ACSTL HR / candidate screening — Chandra Sekhar, Vaishnavi, Indra identified as the active DroneArjuna team.

Evaluation note on the document itself: the plan was authored as an HTML document printed to PDF (browser print-to-PDF, file:///C:/Users/Chandrasekhar/). For future plan documents, authoring in DOCX or the shared document system is preferable to ensure tracked-changes capability and version control compatibility.